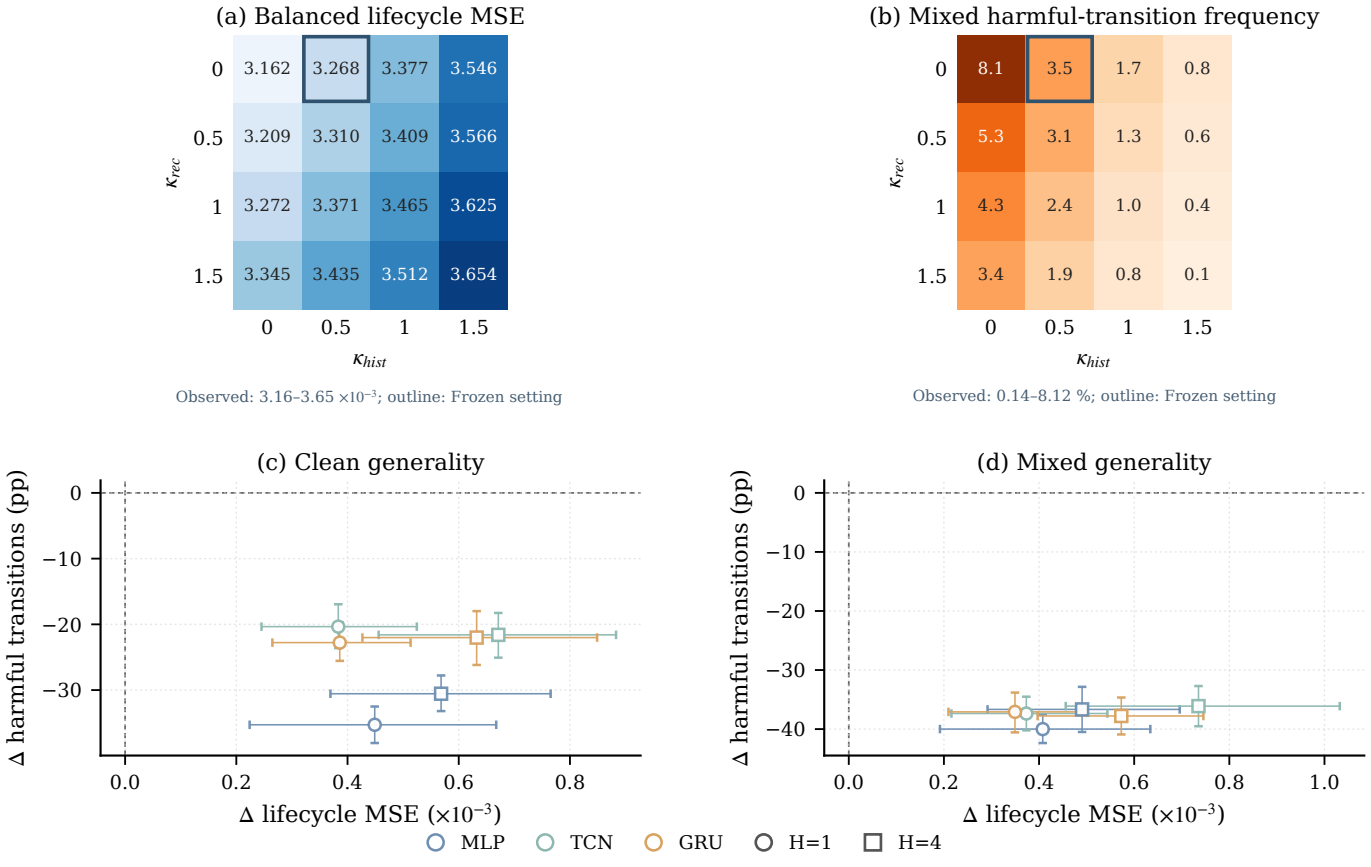
(a) Balanced lifecycle MSE
0
0.5
1
1.5
3.162
3.268
3.377
3.546
3.209
3.310
3.409
3.566
3.272
3.371
3.465
3.625
3.345
3.435
3.512
3.654
κ_{rec}
κ_{hist}
Observed: 3.16–3.65 ×10⁻³; outline: Frozen setting
(b) Mixed harmful-transition frequency
8.1
3.5
1.7
0.8
5.3
3.1
1.3
0.6
4.3
2.4
1.0
0.4
3.4
1.9
0.8
0.1
Observed: 0.14–8.12 %; outline: Frozen setting
(c) Clean generality
Δ harmful transitions (pp)
Δ lifecycle MSE (×10⁻³)
(d) Mixed generality
MLP
TCN
GRU
H=1
H=4